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The Taxation of the Poor

Howard Chernick
Andrew Reschovsky

ABSTRACT

This paper uses microsimulation modeling to estimate the annual burden of federal, state, and local taxes on the poor in two states. We find that in 1988 the average burden of taxation on poor families and individuals was 15.3 percent in Massachusetts and 18 percent in New York. As most of the burden is due to state and local taxes, federal tax reform had only a minor impact on the overall tax burdens faced by the poor. Though the analysis is for two state, we argue that the basic results are applicable for most other states. Given the high incidence of persistent poverty among those who are poor in any given year, we argue that annual burdens provide a good indication of long-run burden for a significant proportion of the poor.

I. Introduction

The tax system should not be an additional burden to those who are struggling to escape from poverty; insofar as possible, those below the poverty line should be freed from taxation altogether.

President's Tax Proposals (1985, p. 5)

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Most commentators on the Tax Reform Act of 1986 have argued that the achievement of this goal, articulated in the above statement from President Reagan's original tax reform proposal is one of the most important features of the new federal tax legislation.[1] By raising the personal exemption, the standard deduction, and the earned income tax credit, the Tax Reform Act of 1986 has reduced the number of poor families paying federal income tax by 4.3 million and eliminated income taxation for most people with incomes below the poverty level (*Economic Report of the President* 1987, p. 85).

But tax reform at the federal level has not been followed at the state and local levels. This paper shows that such taxes continue to place a substantial burden on the poor. Although government transfer programs provide an important source of income for the poor, we do not conduct a net incidence study, but rather focus only on taxes. Using data from the Current Population Survey and a series of linked microsimulation models of federal, state, and local taxes, we develop a comprehensive measure of the tax burden faced by low-income families and individuals. We demonstrate that even before federal tax reform, the largest portion of the tax burden on poor individuals and families originated with state and local, rather than federal, taxes. Consequently, federal tax reform has had a relatively small impact on reducing the overall tax burden faced by many poor families and individuals.

Federal tax reform has also been a catalyst for reexamination of the tax treatment of the poor in several states. A number of states with individual income taxes have responded to the federal changes by increasing personal exemptions and standard deductions, with the objective of removing families in poverty from the tax rolls, and insuring that no taxpayer with zero federal tax liability is obligated to pay state income taxes. According to Hutchinson (1989) in 1989 in 16 of the 40 states with a broad-based personal income tax, a family of four with income below the poverty level was exempted from taxation. At the same time, in 19 other states, families with incomes of less than three-quarters of the poverty line were subject to state income taxation.

To our knowledge, no research has looked directly at the total tax burden faced by persons below the poverty level. In fact, the well-known distributional analyses conducted by Joseph Pechman and his colleagues at the Brookings Institution exclude the poorest 5 percent of families. Microlevel studies of the burden of taxation on the poor have been hampered by the lack of state-by-state data on indirect taxes—sales and ex-

1. For detailed discussions of the federal tax burdens faced by the poor prior to Tax Reform see the following congressional staff reports: U.S. Congress, Joint Committee on Taxation (1984) and U.S. House of Representatives, Committee on Ways and Means (1984).

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cise taxes—and insufficient detail in the modeling of the property tax on renters. To calculate burdens from the indirect taxes it is necessary to estimate the amount of taxable consumption by low-income families living in each state. Under what we believe to be reasonable assumptions concerning property tax incidence (see below), tenants bear a major portion of the real property tax levied on their housing units. Determining property tax burdens for the poor thus requires the capability of modeling the incidence of the property tax on a state-by-state basis. Our approach allows us to estimate tax burdens from both direct and indirect taxes for a random sample of state residents.

Many low-income workers live in families whose incomes exceed the official poverty level by a small amount. In order to include this group, we also analyze the tax burdens faced by those with incomes between the poverty line and 150 percent of the poverty line. This level of income is approximately equal to one-half of median income. For convenience, we will refer to families and individuals with incomes in this range as the "near-poor."

The issue of current versus permanent income is an important one in evaluating the tax burdens of low-income families. Recent research on poverty has emphasized the differences in the nature and duration of spells of poverty for different groups. Our basic measure of incidence is based on current (annual) incomes of the poor. Because the incidence of consumption-based taxes is heavily influenced by the length of the accounting period for income, we also analyze low-income tax burdens according to the extent to which currently poor families are only temporarily poor.

In Section II, we describe our approach to modeling and estimating the tax burdens faced by the poor. We briefly discuss our tax incidence assumptions, outline the structure of the tax simulation models, and describe the income measure used. The following section describes the basic results, focusing on the burdens faced by poor and near-poor residents of two states, Massachusetts and New York, in 1988. This is followed by a discussion of the impact on tax burden calculations of using current rather than permanent income. We next turn our attention to the impact of federal tax reform on the tax burdens faced by the poor. The final section presents our conclusions.

II. Modeling Federal, State, and Local Taxes

In our linked microsimulation models we concentrate on the taxes that are likely to have the largest impact on the poor. At the federal level, we model the individual income tax and the payroll (FICA)

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tax. Together these two taxes account for approximately 85 percent of federal government tax revenue. At the state level we model the two most important revenue sources in most states, the individual income tax and the general sales tax. We also model state and federal excise taxes on alcohol, tobacco, and gasoline. The most important source of tax revenue for most local governments is the property tax. We model the residential component of the local property tax, which is the component with the greatest impact on the poor. The only major tax not included in our analysis is the corporate income tax. There is no consensus within the economics profession concerning its incidence. To the extent that most of the burden of the corporate income tax falls on capital income, it has little impact on the poor. Data from the Current Population Survey indicate that under 10 percent of the poor and the near-poor have any income from interest, dividends, rents, or capital gains; and among those with capital income, capital income averages less than 10 percent of total income. Our choice of incidence assumptions has been guided primarily by the principle that the burden of a tax falls most heavily on the inelastic source or use of income. This principle has led us to make the following assumptions. With respect to the individual income tax at both the federal and state levels, we assume that the tax is borne by the income recipients.[2] Based on evidence of the inelasticity of labor supply, we also assume that the entire burden of the payroll tax falls on workers.[3] We follow the conventional wisdom in assuming that the sales tax is borne by consumers in proportion to their consumption of taxed commodities.[4]

The most controversial issue concerns the property tax, for which there are competing views on incidence. Under the "new view" first articulated by Mieszkowski (1972), the property tax is analyzed as a uniform national tax on capital, coupled with a variable excise component. De-

2. The basis for this assumption is the empirical evidence that workers do not in general change their hours worked, and investors the amount saved, in response to a change in income tax rates. See Pechman (1985) for a brief review of the evidence on labor supply and on saving behavior.

3. An alternative view attempts to link the incidence of the payroll tax to the receipt of social security benefits. Burkhauser and Turner (1985) show that the burden of the tax varies across workers and over time, and that, for those who correctly anticipate their future benefits, the net impact of the payroll tax will always be less than the burden of the tax alone. Their approach seems strongest for workers who are close enough to retirement to correctly anticipate marginal changes in social security benefits, but less relevant for younger workes.

4. See for example, Musgrave, Case, and Leonard (1974) or Pechman and Okner (1974). In several recent papers Edgar Browning (1978, 1985) has argued that sales tax burdens are proportional to factor earnings rather than to consumption expenditures. It is safe to say that Browning's view on sales tax incidence has not gained wide acceptance among public finance economists.

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pending on the model, the excise component is borne by local consumers of housing services, landowners, or labor. For analysts who adopt a national perspective on property tax incidence, the capital tax component dominates the results. From the local perspective the excise component is more important. As we are analyzing property tax incidence in two states, with particular attention to variations in effective rates within the states, this study adopts a local perspective. In so doing, we effectively treat the entire residential property tax as an excise tax, of which the owner-occupied component is borne by the owners of housing and the renter portion divided between tenants and landlords.[5] Studies attempting to combine the national and local components of property tax incidence theory in the study of subnational units of government (De Tray and Fernandez 1986, and Harmon 1989) support our basic finding that low-income households face high residential property tax burdens.[6]

Given the local perspective, the fact that the majority of the poor are renters (75 percent in New York and Massachusetts) makes the incidence of the property tax on rental housing particularly important. In our model the tax on rental property is assumed to be divided between landlord and tenant, with the share of each depending on the share of land in housing value, and the elasticities of supply and demand for housing services. As our base case, we assume 75 percent forward shifting to tenants. We assume that landlords bear the burden of the land portion of the tax, while the improvements component is substantially shifted forward to tenants. Demand for housing, particularly among the poor, is quite inelastic. Though the poor may move frequently, they are restricted in their access to jurisdictions and neighborhoods by fiscal zoning and discrimination (Yinger 1979, 1986). With linear demand and supply curves, the ratio of the change in rent to a change in the property tax is given by $\phi_s/(\phi_s + \phi_d)$, where ϕ_s and ϕ_d are the price elasticities of supply and demand. With a supply elasticity for low income housing of 2.2 (Grieson 1973) and a demand elasticity of -0.4 (Hanushek and Quigley 1980), 85 percent of the tax would be shifted forward to tenants. Our assumption that 75 percent of the property tax is shifted to tenants is thus based on this

5. A similar approach is followed by Ladd and Yinger (1989). Using a multijurisdictional model which allows for the substitutability of land and capital in housing production, and the substitution between housing and other goods in consumption, Case and Grant (1988) find that the excise tax effects of the property tax are large.

6. An alternative approach, associated primarily with Hamilton (1975) treats the property tax as a benefit tax for local public services. This view, which requires strong assumptions regarding zoning or tax capitalization, is most relevant for homogeneous suburban jurisdictions, whereas the poor are increasingly concentrated in central cities. In any case, the explicit purpose of this paper is to analyze tax incidence, independent of the benefits of government expenditures.

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calculation, and on the evidence from Netzer (1966) that land accounts for somewhat less than 25 percent of the value of rental property. The argument for forward shifting is strengthened in cities like New York, where there is significant rent control, and property tax increases are a cost factor in determining allowable rent increases.

The models are constructed as a series of new subroutines to the Transfer Income Model (TRIM), a model developed during the early 1970s by the Urban Institute and the (then) U.S. Department of Health, Education, and Welfare.[7] The results presented in this paper are based on data from the March 1987 Current Population Survey (CPS). The CPS includes information on family structure and size, age, race, sex, ethnic background, and income by source. The income data on the CPS have been inflated to 1988 values using separate state-specific inflation factors for each source of income.[8] A discussion of the income measure is provided below.

Although our models have a great deal in common with several others, in particular, the U.S. Census Bureau (1984) and Urban Institute models (Sulvetta 1976), they have a number of unique features that make them especially powerful as tools for analyzing the tax burdens on the poor. With the exception of the Brookings model (Pechman and Okner 1974) and a model based on aggregate data constructed by Donald Phares (1980), none of the others calculates tax liabilities for all major federal, state, and local taxes. Although in this paper we analyze the tax systems of only two states, Massachusetts and New York, our concentration on these two states has allowed us to model the specific features of these taxes with more precision than is found in other models.

Because tax incidence depends on both the sources and uses of income, and the Current Population Survey contains no information on expenditures, it was necessary to impute certain items to individual filing units. As we explain in more detail in an appendix which is available from the authors, these imputations were made on the basis of behavioral relationships estimated from other data sources. For example, in order to calculate property tax liabilities, we combine data on local property tax rates and assessment ratios, information on within-jurisdiction assessment practices, and data on housing values and rents. We then estimate a statistical relationship between liabilities and a series of economic and

7. For a description of the original TRIM model see Sulvetta (1976). Since its development, the TRIM model has been used to evaluate proposals for a large number of federal government programs.

8. The data for calculating the inflation factors were provided by Data Resources, Inc.

9. For a description of the Census Bureau model see (U.S. Bureau of the Census 1984), and for a description of the Urban Institute model see Sulvetta (1976).

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demographic variables available on the CPS. Our property tax models also carefully determine eligibility for various state and local government programs designed to provide tax relief for low-income and elderly homeowners.

The importance of modeling inter and intrajurisdictional variations in effective property tax rates can be seen by comparing the incidence results to those that would be obtained under a uniform effective tax rate. We estimate that if New York City were able to achieve uniformity in its assessment rates, the regressivity of the residential property tax would be reduced by about 10 percent. In earlier work on the property tax in California (Chernick and Reschovsky 1982), we found that ignoring variations in effective tax rates and bases both within and across jurisdictions results in an approximately 30 percent underestimate of the property tax burdens faced by low-income taxpayers.

The foundation of the sales tax model is a three-stage estimation procedure of taxable consumption. First, data from the Consumer Expenditure Survey (CES) are used to estimate total consumption as a function of income and a series of economic and demographic variables. Next, budget share equations are estimated, where the share of total consumption spent on taxable items is estimated as a function of total consumption and demographic variables. Finally, the estimated shares are multiplied by total consumption to determine taxable consumption in each state.

Our models cannot provide long-run equilibrium responses to specific tax policy changes. Thus, they cannot be described as fully behavioral. However, our models, like the Brookings model and other microsimulation tax models, are designed to allow a wide range of tax incidence assumptions. This flexibility is especially important in modeling state and local taxes.

Our annual income measure includes census money income, an imputed measure of realized capital gains based on IRS data on gains and losses by Adjusted Gross Income class, the cash value of food stamps and federal energy assistance, and the refundable portion of the Earned Income Tax Credit. The only noncash element of income that we include is a crude estimate of the imputed rent from homeownership. Finally, we add the employer's share of the payroll tax to insure that the burden of the payroll tax is invariant with respect to the statutory division of the tax between employer and employee.

Since we do not include the corporation income tax in our basic measure of low-income tax burdens, we do not impute corporation net income to our sample. The exclusion will make little difference in the case of the poor. We also do not include indirect taxes—sales, excise, and property taxes—in income. In a national study of the distribution of the entire tax burden, income valued at factor cost should be reconciled with income

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at product price by including indirect taxes in income. However, any allocation rule for these taxes implies an assumption about the benefits of public expenditures in each state. For example, adding to income an amount equal to the sum of sales, excise, and property taxes implies that all such taxes are strict benefit taxes, for which the market value of services received is exactly equal to the tax borne by each family. As discussed above this assumption is overly restrictive.

We also exclude from income the value of in-kind transfers for medical care and housing assistance. Although a considerable amount of research has been undertaken in trying to develop methods for valuing in-kind transfers, no consensus has been achieved on the best valuation methods. In a recent experimental report, the U.S. Bureau of the Census (1988b) values the average benefits of Medicare and Medicaid coverage at $268 for households in poverty, and places a value of $364 on housing subsidies received by poor households.

The Current Population Survey is known to understate the income levels of poor families, relative to other data sources such as the Michigan Panel Study of Income Dynamics and the Survey of Income and Program Participation. Both as an adjustment for underreporting and as a crude correction for temporarily low income levels, we arbitrarily increased the income of families whose reported income was less than 50 percent of the poverty line to 50 percent of the poverty line. In New York and Massachusetts approximately one-quarter of poor families were affected by this adjustment. As a result their average 1988 incomes increased from $2,045 to $5,565. Even if incomes are accurately reported, current incomes may be low relative to permanent income for some families, making current income an inappropriate basis for computing tax burdens. The permanent income issue is addressed at some length in a later section.

III. Tax Burdens on the Poor

There are about 216,000 families or unrelated individuals below the poverty level in Massachusetts and 932,000 in New York State. (For convenience we shall refer to unrelated individuals as one-person families.) These numbers equal 9.1 percent of all families in Massachusetts and 12.8 percent of all families in New York. Families with incomes between the poverty level and one and a half times the poverty level (the "near-poor") comprise an additional 8.3 and 10.6 percent in Massachusetts and New York, respectively.

The data in Tables 1 and 2 present tax burdens calculated on the basis of 1988 incomes and tax code provisions. Table 1 shows that the average

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Table 1
Tax Liabilities and Tax Burdens of Poor and Near-Poor Massachusetts Families and Unrelated Individuals Based on 1988 Incomes and Tax Law

	Number of Families Paying Taxes[a]	Percent of All Families[a]	Average Tax Liability	Average Tax Burden[b]
Poor Families (Income below the poverty line)				
All Taxes	215,810	100.0%	$1,105	15.3%
Federal income tax				
With positive tax liability	8,150	3.8	−54	−0.5
With negative tax liability	30,750	14.2		
F.I.C.A.	71,310	33.0	210	2.3
State income tax	9,520	4.4	10	0.1
State sales tax	215,810	100.0	180	2.7
Local property tax	158,350	73.4	490	7.0
Excise taxes[c]	215,810	100.0	270	3.8
Near-Poor Families (Income between 100 and 150 percent of the poverty line)				
All taxes	195,490	100.0%	$1,534	12.9%
Federal income tax				
With positive tax liability	47,730	24.4	15	0.3
With negative tax liability	26,790	13.7		
F.I.C.A.	82,900	42.4	531	4.2
State income tax	48,620	24.9	52	0.3
State sales tax	195,490	100.0	195	1.7
Local property tax	139,560	71.4	498	4.2
Excise taxes[c]	195,490	100.0	244	2.2

a. Unrelated individuals are considered to be one-person families.
b. Defined as the average of families' tax liability as a percentage of their total income.
c. Includes state and federal excise taxes on alcohol, tobacco, and motor fuels, plus miscellaneous federal excise taxes.
Source: Massachusetts Tax Simulation Model.

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Table 2
Tax Liabilities and Tax Burdens of Poor and Near-Poor New York Families and Unrelated Individuals Based on 1988 Incomes and Tax Law

	Number of Families Paying Taxes[a]	Percent of All Families[a]	Average Tax Liability	Average Tax Burden[b]
Poor Families (Income below the poverty line)				
All Taxes	931,700	100.0%	$1,216	18.0%
Federal income tax				
With positive tax liability	63,900	6.9	−45	−0.6
With negative tax liability	136,500	14.6		
F.I.C.A.	297,100	31.0	257	2.9
State income tax[c]	27,500	2.9	10	0.1
State sales tax[c]	931,700	100.0	243	3.9
Local property tax	719,400	77.1	553	8.8
Excise taxes[d]	931,700	100.0	198	2.9
Near-Poor Families (Income between 100 and 150 percent of the poverty line)				
All taxes	775,400	100.0%	$2,012	15.4%
Federal income tax				
With positive tax liability	250,200	32.2	97	0.7
With negative tax liability	103,600	13.4		
F.I.C.A.	383,400	49.4	777	5.2
State income tax[c]	270,100	34.8	64	0.4
State sales tax[c]	775,400	100.0	254	2.2
Local property tax	601,400	77.5	621	5.3
Excise taxes[d]	775,400	100.0	199	1.6

a. Unrelated individuals are considered to be one-person families.
b. Defined as the average of families' tax liability as a percentage of their total income.
c. Includes local income and sales taxes.
d. Includes state and federal excise taxes on alcohol, tobacco, and motor fuels, plus miscellaneous federal excise taxes.
Source: New York Tax Simulation Model.

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poor family in Massachusetts faces an annual tax liability of $1,105 and a tax burden (tax liability divided by total income) of 15.3 percent. For the near-poor, the average tax liability and burden are $1,534 and 12.9 percent respectively. Table 2 indicates that in New York State, which has higher state taxes and imposes additional local taxes on sales and income, the average annual tax liability on the poor is $1,216, and the average tax burden is 18.0 percent. The near-poor in New York face an average liability of $2,012 and an average burden of 15.4 percent.

Although in both states all poor families pay some sales tax and about 75 percent face a property tax burden,[10] only one-third of poor families pay payroll taxes, and only a small percentage face positive federal income tax liabilities.[11] Only four percent of Massachusetts families with income below the poverty level must pay state income taxes, and these taxpayers face an average burden of only one percent. In New York fewer than 3 percent of the poor pay any state and local income taxes.

There is considerable variation in the total tax burdens faced by poor families. While under 10 percent of poor families in both states face composite tax burdens of less than 5 percent, about a quarter of all poor face burdens of between 5 and 10 percent, 40 percent face burdens between 10 and 20 percent, and another 30 percent face burdens that exceed 20 percent.

Federal tax reform leaves most poor families and individuals exempt from federal income taxation. Indeed, our data indicate that only 3.8 percent of the poor in Massachusetts and 6.9 percent of the poor in New York face positive income tax liabilities. In all cases it is only unmarried individuals without children who are subject to the income tax. Federal income tax thresholds exceed the poverty level for married couples and single-parent families.

In recent years, payroll (FICA) taxes have been placing an increasing burden on those with earnings. However, only about one-third of poor families in both Massachusetts and New York have any earnings. Among those paying FICA taxes, the average burden is 7.1 and 9.0 percent of income in Massachusetts and New York, respectively.

In order to protect low-income families from high payroll tax burdens, families with dependent children and income from earnings are eligible for the refundable earned income tax credit (EITC). In 1988 eligible tax-

10. There are two reasons why some families face a zero property tax burden. In the case of multiple families living in the same household, we assume that the primary family pays the entire property tax burden. In addition, families that pay no cash rent, or live in public housing are assumed to pay no property tax.

11. For those poor individuals facing positive federal income tax liabilities, the average tax burden is under 2 percent of income in both Massachusetts and New York.

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payers with earnings below $6,225 received a credit that rose with earnings, up to a maximum of $874. For incomes above $9,850, the size of the credit is reduced by ten cents for each additional dollar of income. In both states about 15 percent of poor families receive EITC payments. The average credit equals $438 in Massachusetts and $537 in New York.[12]

How effective is the EITC in reducing federal tax burdens for low-income families? In both Massachusetts and New York the average credit going to families below the poverty line is large enough to completely offset both the federal income tax and the employee share of the payroll tax. For the near-poor the effectiveness of the EITC is greater for the single-parent than for two-parent families, and for families with fewer children. In Massachusetts, 55 percent of single-parent families are relieved of direct federal taxes, while only 10 percent of two-parent families have an EITC large enough to outweigh FICA and federal income tax liabilities. In New York less than a quarter of all near-poor families have their liabilities offset.

Two-parent families tend to face higher tax burdens than single-parent families because they generally have higher levels of earnings.[13] In New York the average two-parent family classified as near-poor faces a combined federal income and payroll tax burden of nearly 6 percent, versus 3 percent for single parent families. Controlling for poverty status, families with more children also tend to have higher earnings than smaller families. As the EITC by design is independent of family size, larger families in general face higher burdens from federal income and payroll taxes.

The above results refer to EITC-eligible families. Slightly more than half of all poor and near-poor families and individuals with earnings have no children and hence are not eligible for the EITC. On average these taxpayers face the heaviest burdens from federal taxation. This group of working poor and near-poor face combined FICA and federal income tax burdens of about 6 percent in both Massachusetts and New York.

Despite the fact that some low-income people pay federal taxes, it is important to keep in mind that in Massachusetts and New York, two-thirds of all poor families and about half of near-poor families have no earnings and hence pay no FICA and (in most cases) federal income taxes. Thus, for this group, the entire tax burden comes from state and

12. In a conversation with Richard Kasten of the Congressional Budget Office, he indicated that a comparison of the actual number of EITC recipients with a Census-based estimate of the number of eligibles showed that at least 80 percent of those eligible for the EITC do in fact take advantage of the credit.

13. As consumption patterns are similar for single-parent and intact low-income families, these two types of families face similar burdens from state and local taxes.

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local taxes. Even for those low-income families and individuals who have positive federal tax liabilities, state and local taxes average over 60 percent of total taxes.

Undoubtedly influenced by recent public attention focusing on the federal income taxes paid by the poor, a number of states have taken explicit steps to reduce state income tax burdens on their poor residents. Although both New York and Massachusetts place a relatively heavy reliance on the individual income taxes, both states have recently taken action to eliminate state income tax liabilities on the poor. In Massachusetts, effective in 1987, the definition of "no tax status" has been expanded so that no tax is due on adjusted gross income below $8,000 ($12,000 for couples filing joint returns). These no-tax levels are above the poverty line for all individuals and families except for single parents and couples with more than two dependents. In New York, by 1987 the sum of personal exemptions and the standard deduction exceeded the poverty level for most taxpayers.

The data in Tables 1 and 2 show that poor families in Massachusetts pay 2.7 percent of their income in general sales taxes and in New York almost four percent. These burdens occur despite the fact that the sales tax is relatively unimportant in both states, and both states exempt from taxation the purchase of most goods and services generally considered to be necessities. The high sales tax burdens for low-income families occur primarily because, despite the intentions of state government policymakers, taxable consumption as a proportion of total consumption does not vary significantly across income levels. We estimate that the elasticity of taxable consumption as a proportion of total consumption with respect to total income is not significantly different from zero.

To understand why the attempt to exempt purchases of necessities from sales taxation is not more successful in reducing sales tax burdens on the poor, it is useful to look in more detail at the spending pattern of low-income families. Data from the Diary Survey of the 1985 Consumer Expenditure Survey indicate that among the poorest 20 percent of all families, nearly 30 percent of food expenditures are made on food consumed away from home, and are thus subject to sales taxation in most states (U.S. Bureau of Labor Satistics 1987). National data from the Diary Survey indicate that adding consumption expenditures on food consumed away from home to just three categories of taxable expenditures—personal care products, nonprescription drugs and supplies, and housekeeping supplies—accounts for almost 20 percent of the total income of the poorest one-fifth of all U.S. families. Among families with incomes below $5,000 these categories of taxable expenditures account for over 35 percent of total income. Applying the median state sales tax

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rate of 5 percent to these expenditures yields a tax burden of about 1.7 percent of income. Thus, even though most states exempt expenditures on a broad range of goods and services, the poor continue to face relatively highs sales tax burdens in most states.

In both Massachusetts and New York the largest source of the tax burden on the poor comes from the property tax. Among those facing a property tax burden, the average tax liability is $668 in Massachusetts and $716 in New York, and the average burdens are 9.5 and 11.4 percent, respectively. Burdens on owners substantially exceed those on renters in Massachusetts, but are about equal in New York. Over 60 percent of poor families paying property taxes are tenants. In Massachusetts they face an average property tax burden of 7.9 percent of income, while poor homeowners face burdens that average 12.1 percent.

The inclusion of imputed rent in total income has a substantial impact on the property tax burdens of low-income homeowners. Imputed rent comprises about 40 percent of total income for poor owners, and in both states the average property tax burden for this group would rise by 65 percent or more if imputed rent were excluded from the income measure. For the near-poor owner, the average burden would rise by about 50 percent—from 7.8 to 12.1 percent in Massachusetts, and 7.1 to 10.6 percent in New York. Because imputed rent comprises almost half of total income for poor elderly homeowners, if it is not included in income their measured property tax burden would nearly double.

Although an appropriate measure of "ability to pay" should include imputed rents from homeownership, it should be emphasized that property taxation can create substantial economic hardship for elderly homeowners who, despite large amounts of home equity, have few liquid resources with which to pay their property taxes and provide for the other necessities of life. Although reverse-equity mortgages and tax deferral programs operated in some states may potentially increase the liquidity of home equity, most homeowners appear to be either unable or unwilling to participate in these programs.

Property tax burdens faced by poor tenants are high enough to cause economic hardship for many poor tenants. The data in Tables 1 and 2 are based on the assumption that tenants bear 75 percent of the total property tax burden on their housing unit. Although this assumption is based on a careful assessment of the low-income rental housing market, it is possible that landlords are able to shift forward to tenants either a smaller or larger share of the total burden on rental property. We thus conducted a sensitivity analysis of the forward shifting assumption by considering both 50 percent and 100 percent shifting of the tax burden. The possibility of complete forward shifting to tenants is most likely in areas, especially

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cities, where exceedingly high rent burdens on low-income tenants indicate a very tight market for rental housing.[14] The results of this sensitivity analysis show that tenant property tax burdens would range from 5.3 to 10.5 percent in Massachusetts and from 7.6 to 15 percent in New York under these two alternative shifting assumptions. The composite tax burden on tenants correspondingly ranges from about 13.5 to 19 percent in Massachusetts and 14 to 21.5 percent in New York.

IV. Persistent Poverty and the Burden of Taxation

One criticism of tax-incidence computations based on cross-sectional data is that they overstate the burdens of those with low incomes (Pechman 1985, Davies, St.-Hilaire, and Whalley 1984). This is particularly true for taxes whose incidence is determined primarily by the uses of income, as we assume to be the case for the sales, excise, and residential property taxes. Analysts typically ascribe the high annual tax burdens for the lowest income classes to the large negative transitory component of measured income and frequently exclude the very poor from the analysis. Since consumption is more nearly proportional to permanent income than to annual income, the average tax burden for consumption tax borne by low-income families will be lower as the accounting period is lengthened.

Our imputed consumption measures are based on estimated consumption relationships (from the Survey of Income and Education for the property tax and the CES for sales and excise taxes) as a function of reported annual income and demographic characteristics. The ratio of consumption expenditures to income in the CES typically exceeds three in the lowest quintile of the income distribution and falls sharply as income rises. Since the estimated income elasiticites of demand are biased downward, there is an upward bias in the estimated consumption tax burdens of families whose annual incomes are low.

One approach to the permanent income problem is to attempt to assess the degree of bias in our marginal propensity to consume estimates by

14. Sandra Newman and Ann Schnare (1986) show, using evidence from the 1983 American Housing Survey, that in 1983 of the 8.5 million poor households not receiving public assistance (about one-half of all poor households), some 60 percent paid more than half their income in rent, while of the 4.6 million households receiving income but not housing assistance, the average rent burden was almost 50 percent of income. The likelihood of complete forward shifting is probably greater in Boston and New York City because both these cities have rent control ordinances which allow property tax increases to be fully reflected in higher rents.

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reference to other studies.[15] Though in general the bias in using reported annual income depends on the ratio of the variance of transitory income to the variance of permanent income, the degree of bias should vary by consumption category. Mayo (1981) reports that there is a significant downward bias from using current income in housing demand estimates—from 16 percent to 50 percent—and it is greater for renters than for owners. Comparable estimates for taxable consumption under sales and excise taxes are not available, though in general one might expect the bias to be lower, because it is less costly to adjust consumption levels of these items than to adjust housing outlays in response to transitory reductions in income. A rough approximation of the upper-bound of the sales tax bias can be inferred from a recent estimate of the overall marginal propensity to consume using CES consumption data but an instrument for income (Clayton-Matthews and Kazarosian 1988). The estimated sales tax burden for poor families in New York State would decline from 3.9 percent to about 1.5 percent under this estimate.[16]

The upward bias in the sales and excise tax estimates for the poor from using reported income is offset to an unknown degree by the general tendency toward underreporting of expenditures on the CES. As a result of this underreporting, simulated sales tax liabilities understate the household portion of the sales tax by about 50 percent in both states.[17]

An alternative approach to the permanent income problem is to distinguish between temporary and persistent poverty, and focus on the tax burdens of those who are likely to be poor for an extended time period. Recent literature on income mobility reports a substantial degree of instability in the overall income distribution.[18] Panel studies of poverty support this conclusion, indicating relatively short durations for most spells of

15. By contrast, Poterba (1989) simply assumes that consumption is proportional to lifetime income, and that average annual lifetime income can be approximated by annual consumption. Thus, tax burdens are equal to the share in total consumption of the taxable item or items.

16. The estimate in the text is derived as follows. Clayton-Matthews and Kazarosian (1988) estimate a total expenditure equation from the 1985 CES, using an instrument for disposable income. Under their equation consumption expenditures at an income level of $5,000 would be almost $5,000, rather than the substantially higher predicted consumption if reported income is used. Our estimates from the CES show that on average taxable consumption is equal to about 20 percent of total consumption, and that the budget share does not vary significantly by income level. Applying the New York tax rate to estimated taxable consumption of $1,000 yields a tax burden of 1.5 percent.

17. A similar type of underestimate for the sales tax in all states is reported by Feenberg and Rosen (1986).

18. Poterba (1989) summarizes a number of studies of income dynamics which report relatively high degrees of income mobility.

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poverty (Ruggles and Williams 1988). At the same time, a number of studies have emphasized the issue of persistent poverty.

Papers by Levy (1977), Gottschalk (1982), and Bane and Ellwood (1986) indicate that, while spells of poverty are temporary for many of those who fall below the poverty line, a high proportion of the poor at any given time are likely to be poor over an extended time period. For example, Bane and Ellwood report that "more than 50 percent of . . . the poor in a cross-sectional survey are in the midst of the spell of poverty which will last ten years or more" (p. 11). These findings suggest that even though the negative transitory component may be a significant fraction of annual income for the poor as a group, for a substantial subset annual income may provide a reasonable approximation to average income over a longer period.

The consumption tax incidence results presented in the previous section of the paper were obtained by dividing predicted consumption by an income measure that was adjusted upward by the arbitrary inflation of very low reported incomes and by the inclusion of imputed income from owner-occupied housing.

A second approach to the approximation of long-run tax burdens is to apply differential weights to tax burdens according to the probability that the family is in the midst of a long-run spell of poverty. The permanence of poverty will, of course, vary with the criterion used. For the purpose of evaluating tax burdens, we argue that a relatively less stringent definition of permanence is appropriate—i.e., incomes below the poverty line in at least five years out of a ten-year period—rather than the ten-year duration period cited in the Bane and Ellwood study. The five-in-ten criterion can be justified by the fact that for most such families incomes are unlikely to rise very far above the poverty line even over a relatively long accounting period.[19]

To determine the incidence of persistent poverty in our sample, we draw on recent estimates by Adams and Duncan (1988) of the correlates of persistent poverty among urban residents. Based on a ten-year sample from the Panel Study of Income Dynamics, they find that race and sex of head are significant predictors of persistent poverty. The probability that a family falls below the poverty line in at least five out of ten years ranges from 50 percent for nonblack male-headed families to over 75

19. Ruggles and Williams (1988) find that even for entrants into spells of poverty as short as a month, about half have two-year average income levels below 150 percent of the poverty line, and about three-quarters have average incomes below 200 percent of the poverty line. This finding of low longer-term incomes is expected to be even stronger for the group of persistently poor we are examining in this section of the paper.

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percent for those whose head is black and female.[20] Applying these correlates to the New York sample, two-thirds of the poverty population is likely to be poor over an extended time period.

Permanent income theory suggests that the persistently poor are more likely to have adjusted their consumption levels to reflect their lower long-run incomes, and therefore will tend to face lower property and sales tax burdens than the temporarily poor. To identify those likely to be persistently poor, we divided families by race and sex of head, and ranked each group from low to high according to the combined burden of sales, property, and excise taxes. In this exercise, income was not adjusted upward and did not include imputed rent. Using New York data, the median burden of those poor likely to be persistently poor was 10.6 percent. This compares to an average burden of consumption taxes for all the poor in New York of 15.6 percent (see Table 2). These results suggest that even for those likely to be poor for an extended time period, the burden of consumption taxes is still significant. For the temporarily poor, whose incomes may not rise very far above the poverty line even when they exit from poverty (see footnote 19), annual burdens are more than 25 percent of income.

Since taxes are paid out of current income, high tax burdens are a significant policy issue, whatever the time period. With cross-sectional data, we cannot adequately measure the extent to which high burdens persist over time. However, the above discussion suggests that the high annual burdens predicted by our models may be good approximations of long-term tax burdens for a significant proportion of the poor.

V. The Impact of Federal Tax Reform

It is important to put the impact of federal tax reform in perspective. Although federal tax reform reduced the number of poor paying federal income taxes by over 60 percent and the number of near-poor paying income taxes by about one-quarter, the large majority of the poor and near-poor were not directly affected by federal tax reform.[21] If the Tax Reform Act of 1986 had not been enacted, in 1988 close to 80

20. These proportions are derived from Table 6 of Adams and Duncan (1988). The authors also calculate that at least 4 percent of nonpoor in 1979 are likely to be persistently poor in 1974–1983.

21. These results were derived by comparing a calendar year 1988 simulation under current law with a simulation based on federal income tax law as it would have been in 1988 in the absence of tax reform.

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percent of poor families would not have been liable for federal income taxes (or eligible for refundable credits). Even among those who paid federal income taxes prior to federal tax reform, the total tax burdens from all taxes was only reduced by about three percentage points in both Massachusetts and New York. For most low-income families the burden of taxation continues to come primarily from state and local taxation.

By expanding the earned income tax credit, federal tax reform offsets social security taxes for an increased number of low-income families. Federal tax reform results in an increase of nearly 35 percent in the number of poor and near-poor families receiving EITC credits that exceed their FICA tax liabilities. It should be recalled, however, that even after federal tax reform, about two-thirds of the poor with earnings and over 80 percent of the near-poor with earnings continue to face FICA (and in some cases federal income tax) liabilities that are not offset by federal credits.

By raising personal exemptions, the standard deduction, and the EITC, federal tax reform has restored the income tax threshold of most families to a level above the poverty line for the first time since 1980. During the 1980s, when most moderate and high-income taxpayers were benefiting from successive tax cuts, more and more low-income taxpayers were being forced to pay federal income taxes for the first time, as inflation reduced the real value of personal exemptions, the standard deduction, and the EITC. In effect, federal tax reform has merely returned most poor families to the position they held in 1980 relative to the income tax.

Although the indexing of the major features of the income tax will prevent the income tax threshold from slipping below the poverty level, the increase in the combined FICA tax rate to 15.3 percent in 1990 will further increase federal tax burdens on many poor and near-poor taxpayers.

VI. Conclusion

In this paper we set out to see whether recent tax reform efforts at the federal, state, and local levels have been effective in alleviating tax burdens on the poor. Although the taxation of the poor is frequently discussed, both conceptual and practical measurement problems have resulted in few attempts to determine the overall tax burdens faced by poor individuals and families. Using a microsimulation model of the major federal, state, and local personal taxes, we have shown that low-income families in New York and Massachusetts face tax burdens of 18 percent of income in New York and 15 percent in Massachusetts. The

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major sources of these high tax burdens are state and local sales and property taxes, and federal and state excise taxes. Because relatively few poor families paid federal income taxes even before tax reform, the Federal Tax Reform Act of 1986 was largely ineffective in reducing the total tax burden faced by the poor.

Because of the important contribution of taxes on consumption to the overall tax burden of the poor, we have paid special attention to the potential upward bias in tax burdens caused by the use of annual rather than permanent income. When we weight the estimated burden of consumption taxes according to the likelihood of persistent poverty, we find that in New York the median tax burden for the persistently poor is estimated to be 10.6 percent, as compared to 15.3 percent for the entire sample. Moreover, since most of those experiencing even temporary spells of poverty do not rise very far above the poverty line over longer time periods, the high estimated consumption tax burdens (over 25 percent of income) for the nonpersistently poor in our sample should not be completely discounted. Therefore, we argue that for the majority of poor families, high tax burdens are not simply artifacts of the annual accounting period, but indicate significant real welfare burdens.

A question that arises in the interpretation of any empirical incidence study is the sensitivity of the results to the incidence assumptions and the data employed. Because there is no consensus on property tax incidence assumptions, we have simulated the impact of several alternative assumptions. First, in line with the "new view" of property tax incidence, we have assumed that 40 percent of the burden is equivalent to a national property tax and is allocated to the owners of capital. The result of this assumption is to reduce the average property tax burden on the poor by about 30 percent, and the overall tax burden on the poor by about 15 percent. For reasons outlined earlier, the allocation of most of the property tax burden to the consumers of housing is our preferred incidence assumption for low-income families. Second, we have considered alternative forward-shifting assumptions for the property tax on rental housing. Assumptions of 50 and 100 percent shifting yielded a range of property tax burden estimates for renters from 5 to 10 percent in Massachusetts and 8 to 15 percent in New York. However, recent data on rent burdens for low-income households suggest that the upper-bound estimates may be more accurate than the lower-bound.

Although this study is based on data from only two states, we believe that the basic results hold for residents of most states. Massachusetts and New York are different from the average state in several ways, but the impact of these differences on tax burdens faced by the poor tend to be offsetting. In both states the proportion of the poor with earnings is

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lower than in the average state,[22] implying a reduction in the average FICA tax burden on the poor, but also a smaller role for tax reform in reducing federal tax burdens on the poor than in other states.

New York is clearly a "high tax" state, with state and local taxes as a percent of personal income more than 40 percent above the national average. Massachusetts is more typical of the nation, with a burden less than 4 percent above the national average in fiscal year 1987 (U.S. Bureau of the Census 1988a).[23] However, there are several aspects of the tax structure in the two states that tend to mitigate the resulting higher than average tax burdens on the poor.

First, both states place above-average reliance on the personal income tax as compared to taxes on consumption. However, while most states levy income taxes on the poor, both Massachusetts and New York exempt the poor and the majority of the near-poor from state income taxation. Second, exemptions of necessities makes the sales tax less regressive than in most other states. Using Suits indices, Donald Phares (1980) found that in 1975 only one state had a sales tax that was less regressive than Massachusetts's and only seven states had sales taxes less regressive than New York's. Third, both states levy much lower excise taxes on motor fuels than most other states. New York collects less revenue from the tax (in both per capita terms and as a percentage of personal income) than all other states, and Massachusetts ranks very near the bottom (Advisory Commission on Intergovernmental Relations 1988).

The large federal budget deficit and recent revenue shortfalls in a number of states have led many economists (and some political leaders) to propose increasing federal and state excise tax rates on gasoline, cigarettes, and alcoholic beverages. Our analysis indicates that a substantial portion of the tax burden on the poor (about 25 percent in Massachusetts and 16 percent in New York) currently comes from these excise taxes. Further increases in these taxes without offsetting measures to protect low-income taxpayers will only serve to increase the economic hardships faced by most low-income families.

Our results suggest that even after the passage of the Federal Tax Reform Act of 1986, the burden of taxation on the poor remains high in absolute terms. Our analysis is not a net incidence study, but focuses only on taxes. Because government transfer programs provide an important source of income, both cash and in-kind, for the poor, a general assessment of antipoverty policy would include both taxes paid and bene-

22. It is interesting to note that CPS data indicate that in 1984 both states also had a lower than average proportion of poor who had any earnings.

23. In per capita terms, state and local taxes in Massachusetts were 26 percent above the national average in fiscal year 1987.

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fits received. We believe, nonetheless, that our results support the case for further tax relief for low-income families.

Recently proposed federal legislation to expand the Earned Income Tax Credit and to convert the federal child care credit into a refundable credit provide one way to protect low-income families, especially those with earnings, from further increases in tax burdens. Similar steps at the state level, combined with increases in property tax relief measures targeted to low-income tenants and homeowners could provide substantial tax relief from existing burdens.

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